

Disproof of conjecture 00000008234

earthking11

2026-09-13

Verdict: FALSE

We disprove conjecture 00000008234 as stated. The conjecture is a conjunction of several clauses, the first of which asserts that for an arbitrary number of agents dividing an arbitrary number of indivisible goods, the EF1 exchange graph is connected with diameter at most $m(n - 1)$ under arbitrary monotone valuations. We exhibit an elementary counterexample with $n = 2$ agents, $m = 2$ goods and unit valuations: the EF1 exchange graph has exactly two vertices and *no* edge, so it is disconnected and its diameter is infinite. The claimed bound $m(n - 1) = 2$ therefore fails, and the first conjunct of the conjecture is false; hence the whole conjunction is false.

1 The conjecture

Quoted verbatim from `conjectures/00000008234.md`:

Definition: The exchange graph `swap_graph` of envy-free-up-to-one-good allocations when n agents divide m indivisible goods: vertices are EF1 allocations and edges are single-good transfers. **Conjecture:** Under arbitrary monotone valuations `swap_graph` is connected with diameter at most $m(n - 1)$; an allocation that is simultaneously EF1 and Pareto optimal always exists and is reachable by a polynomial-time item-moving algorithm; and EFX allocations always exist for three agents, while a minimal counterexample, if one exists, must have exactly 4 agents and require a non-bimodal valuation profile.

We refute the first conjunct. Throughout, `swap_graph` has as vertices the EF1 allocations and as edges the unordered pairs of EF1 allocations that differ by a single-good transfer.

2 Allocations, unit valuations, and EF1

Definition 1 (allocation). An allocation of m indivisible goods to n agents is a map $a : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ sending each good to its owner. The bundle of agent i is $A_i = a^{-1}(i)$.

Definition 2 (unit additive valuation). The *unit* valuation assigns every agent the same value $v_i(g) = 1$ to every good g , extended additively to bundles: $v_i(A) = |A \cap A_i|$ for the bundle A_i of agent i . Unit valuations are additive, hence monotone: if $A \subseteq B$ then $v_i(A) \leq v_i(B)$. Thus they are admissible under the conjecture's hypothesis of arbitrary monotone valuations.

Definition 3 (EF1). An allocation a is *envy-free up to one good* (EF1) if for every ordered pair of

agents (i, j) there is a good $g \in A_j$ such that

$$v_i(A_j \setminus \{g\}) \leq v_i(A_i).$$

Equivalently, whenever i envies j (that is, $v_i(A_j) > v_i(A_i)$), the removal of a single suitable good from A_j eliminates the envy.

Lemma 4 (unit-valuation characterisation). *Under unit valuations, an allocation a is EF1 if and only if*

$$\max_i |A_i| - \min_i |A_i| \leq 1.$$

Proof. Write $s_i = v_i(A_i) = |A_i|$. ($\Leftarrow$) Let i, j be agents. If $s_i \geq s_j$ then i does not envy j , and the EF1 condition is satisfied. Otherwise $s_j \geq s_i + 1$. If $s_j = s_i + 1$ then $A_j \neq \emptyset$ and removing any $g \in A_j$ gives $v_i(A_j \setminus \{g\}) = s_j - 1 = s_i \leq s_i$, so the condition holds. If $s_j \leq s_i + 1$ for all i, j , one of these two cases applies to every ordered pair. ($\Rightarrow$) Suppose $s_j \geq s_i + 2$ for some i, j . For every good $g \in A_j$ we have $v_i(A_j \setminus \{g\}) = s_j - 1 \geq s_i + 1 > s_i$, so no single removal eliminates the envy of i towards j . Hence a is not EF1. Contrapositively, EF1 forces $s_j \leq s_i + 1$ for all ordered pairs, i.e. $\max_i s_i - \min_i s_i \leq 1$. $\square$

3 The counterexample: $n = 2$ agents, $m = 2$ goods

Take goods $\{g_1, g_2\}$ and agents $\{0, 1\}$ with unit valuations. There are $2^2 = 4$ allocations. By Lemma 4, EF1 is equivalent to the bundle sizes differing by at most 1. The complete table is:

allocation a	bundles (A_0, A_1)	sizes (s_0, s_1)	EF1?
both goods to agent 0	$(\{g_1, g_2\}, \emptyset)$	$(2, 0)$	no
$g_1 \mapsto 0, g_2 \mapsto 1$	$(\{g_1\}, \{g_2\})$	$(1, 1)$	yes
$g_1 \mapsto 1, g_2 \mapsto 0$	$(\{g_2\}, \{g_1\})$	$(1, 1)$	yes
both goods to agent 1	$(\emptyset, \{g_1, g_2\})$	$(0, 2)$	no

Theorem 5. *For $n = m = 2$ with unit valuations there are exactly two EF1 allocations, namely the two diagonal allocations*

$$a_{01} : g_1 \mapsto 0, g_2 \mapsto 1, \quad a_{10} : g_1 \mapsto 1, g_2 \mapsto 0,$$

and no others.

Proof. By the table, the allocations with size vector $(2, 0)$ and $(0, 2)$ have a gap of 2 and are not EF1 by Lemma 4. The two allocations with size vector $(1, 1)$ have gap $0 \leq 1$ and are EF1. The four allocations are all of them, since each of the two goods has two possible owners. Hence exactly a_{01} and a_{10} are EF1. $\square$

Theorem 6. *In the above instance, swap_graph has no edge. Consequently swap_graph is disconnected and has infinite diameter; in particular the conjectured bound $m(n - 1) = 2$ fails.*

Proof. The vertices of swap_graph are a_{01} and a_{10} by Theorem 5. They differ in both coordinates, so their Hamming distance is 2; they are not related by a single-good transfer.

It remains to check that a single-good transfer out of either vertex does not land on a vertex. Starting from a_{01} , transferring g_1 to agent 1 yields the allocation with sizes $(0, 2)$, and transferring

g_2 to agent 0 yields the allocation with sizes $(2, 0)$; both have size gap 2 and are not EF1 by Lemma 4. The same two targets arise from a_{10} . Hence no single-good transfer has both endpoints EF1, i.e. swap_graph has no edge.

A graph on two vertices with no edge has two connected components, so it is not connected, and its diameter is infinite (there is no path, so no finite bound on path lengths). Since the conjecture asserts connectivity with diameter at most $m(n - 1) = 2$, its first conjunct is false. $\square$

Remark 7. The argument is even stronger than disconnection: *every* single-good transfer out of an EF1 allocation leaves the EF1 set. Thus swap_graph is a totally disconnected graph on two vertices, and the same happens in every instance of the next section.

4 An infinite family: n divides m

The phenomenon is not an artifact of the smallest instance.

Theorem 8. *Let $n \geq 2$ and $m \geq n$ with $n \mid m$, and use unit valuations. Then every EF1 allocation is balanced, i.e. $|A_i| = m/n$ for every agent i . Consequently every single-good transfer produces an allocation that is not EF1, so swap_graph has no edges at all. Since there are at least two balanced allocations, swap_graph is disconnected and its diameter is infinite.*

Proof. Let a be EF1 and put $s_i = |A_i|$. By Lemma 4, $\max_i s_i - \min_i s_i \leq 1$, so there is an integer k with $s_i \in \{k, k + 1\}$ for all i . Let r be the number of agents with $s_i = k + 1$; then

$$m = \sum_{i=1}^n s_i = nk + r, \quad 0 \leq r \leq n.$$

Since $n \mid m$, we have $r \equiv 0 \pmod{n}$, so $r = 0$ or $r = n$, and in both cases all s_i are equal, with common value m/n . (Equivalently: an average that is an integer and all sizes within 1 of one another forces all sizes to equal the average.)

Now perform one single-good transfer: some agent i loses a good and some agent j gains one, so the new sizes are $m/n - 1$ for i and $m/n + 1$ for j , while every other agent still has m/n . The gap is $(m/n + 1) - (m/n - 1) = 2 > 1$, so by Lemma 4 the new allocation is not EF1. As every EF1 allocation has every agent at exactly m/n goods, no single-good transfer has both endpoints EF1, and swap_graph has no edges.

Finally, the number of balanced allocations is the multinomial $m!/((m/n)!^n)$, which is at least 2 when $n \geq 2$ and $m \geq n$ (for $m = n$ it is $n! \geq 2$, and for $m/n \geq 2$ there are again at least two ways to choose one agent's share). With at least two vertices and no edge, swap_graph is disconnected. $\square$

Corollary 9 ($n = m = 3$). *Three agents and three goods with unit valuations form a counterexample with three (not two) agents. The EF1 allocations are exactly the six permutations that give each agent one good; all pairwise Hamming distances are at least 2, so swap_graph has no edges and is disconnected.*

Proof. By Theorem 8 with $n = m = 3$, every EF1 allocation assigns exactly $3/3 = 1$ good to each agent. There are $3! = 6$ such allocations. Two distinct permutations differ in at least two positions, so their Hamming distance is at least 2; a single-good transfer moves exactly one good, landing on

a size vector with a gap of 2, which is not EF1. Hence no edge, and the six vertices split into six components. $\square$

The following table records the disconnection for several members of the family (verified by exhaustive enumeration in `reproduce.py`):

n	m	# EF1 allocations	# edges
2	2	2	0
2	4	6	0
2	6	20	0
3	3	6	0
4	4	24	0

5 Conclusion

With $n = 2$ agents, $m = 2$ goods and unit (hence monotone, additive) valuations, the EF1 exchange graph `swap_graph` has exactly two vertices and no edges. It is disconnected, its diameter is infinite, and the conjectured bound $m(n - 1) = 2$ fails. The same holds for every $n \mid m$ with $n \geq 2$. This refutes the first conjunct of conjecture 00000008234, and therefore the conjecture as a conjunction.

6 Reproducibility and formalisation

The exhaustive enumeration, the literal EF1 predicate, the swap graph and its connected components are computed by `reproduce.py` (standard-library Python 3 only), which prints the tables above, asserts zero edges and disconnection, and ends with `PASS`.

The 2×2 refutation is formalised in Lean 4 in the accompanying `lean4/` project, core Lean only (no Mathlib, no `sorry`). The main formal statements are

```
theorem exactly_two_ef1 : ef1 alloc01 ∧ ef1 alloc10 ∧ ...
theorem no_adj_ef1 : ef1 a → ef1 b → ¬ Adj a b
theorem not_ef1_connected : ¬ EF1Connected
theorem no_path_between_diagonals : ¬ (∃ k, ReachN alloc01 k alloc10)
theorem claimed_diameter_bound_fails : ¬ SwapGraphDiameterAtMost 2
```

Here `ef1` is the EF1 predicate transcribed literally (for every ordered pair of agents, some good can be removed from the envied bundle to remove the envy), `Adj` is Hamming distance 1 (a single-good transfer), `ReachN` is reachability through EF1 vertices, and `SwapGraphDiameterAtMost` is the conjectured connectivity plus diameter bound. `Check.lean` prints the axioms of each theorem, confirming that only `propext` and `Quot.sound` are used and that `sorryAx` does not occur.